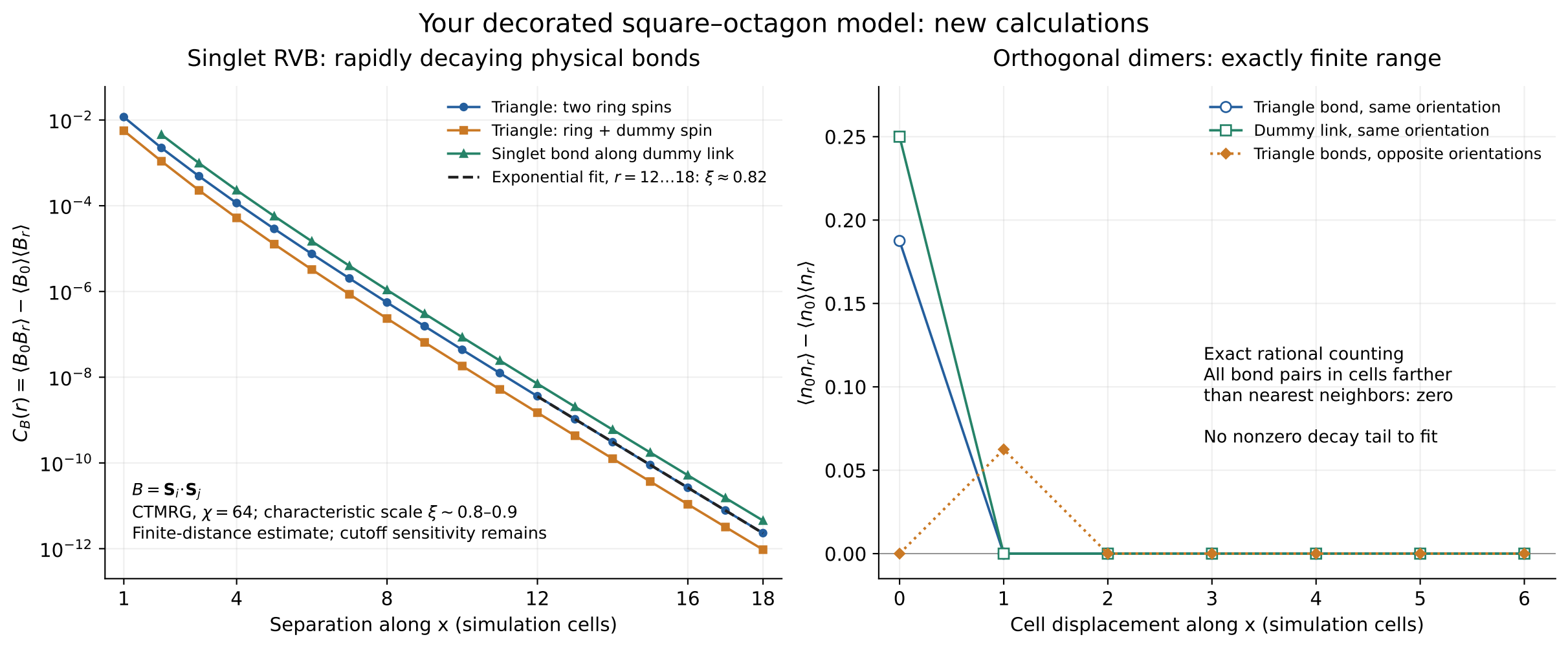
Your decorated square–octagon model: new calculations
Singlet RVB: rapidly decaying physical bonds
Orthogonal dimers: exactly finite range
Triangle: two ring spins
Triangle: ring + dummy spin
Singlet bond along dummy link
Exponential fit, r = 12…18: ξ ≈ 0.82
$C_B(r) = \langle B_0 B_r \rangle - \langle B_0 \rangle \langle B_r \rangle$
$B = \mathbf{S}_i \cdot \mathbf{S}_j$
CTMRG, χ = 64; characteristic scale ξ ~ 0.8–0.9
Finite-distance estimate; cutoff sensitivity remains
Separation along x (simulation cells)
Triangle bond, same orientation
Dummy link, same orientation
Triangle bonds, opposite orientations
$\langle n_0 n_r \rangle - \langle n_0 \rangle \langle n_r \rangle$
Exact rational counting
All bond pairs in cells farther
than nearest neighbors: zero
No nonzero decay tail to fit
Cell displacement along x (simulation cells)